A Effects Safety Theorem

Follow the proofs of System F^{ε} in [?], our language extends a new type blab, and $\texttt{error}_p$ for migrating labels of Contracts to Effects. We also extend with shallow handler which is very similar to deep handler. We also add recursive function μ. These are the extra details that we must extend the proofs.

First we redefine some "operations" and rules that change when when adding the above.

- $\lceil E \rceil^l$ extracts all labels l from a context in reverse order:

  $\lceil F_0 \cdot \texttt{handle}_m(h_1, \cdot) \cdot F_1 \cdot ... \cdot \texttt{handle}_m(h_n, \cdot) \cdot F_n \rceil^l \triangleq \langle l_n, ..., l_1 \rangle$ iff $h_i : \Sigma(l_i)$

  Adapted to include both shallow and deep handle.

- Evaluation context typing

  Adapted to include both shallow and deep handle.

$$\frac{\Gamma \vdash_{\text{ops}} m : h : \sigma \mid l \mid \varepsilon \qquad \Gamma \vdash_{\text{ec}} E : \sigma_1 \to \sigma \mid \langle l, \varepsilon \rangle}{\Gamma \vdash_{\text{ec}} \texttt{handle}^{\varepsilon}_m(h, E) : \sigma_1 \to \sigma \mid \varepsilon} \text{CHANDLE}$$

A.1 Evaluation Context Typing Lemmas

Lemma A.1. (Evaluation Context Typing)

If $\Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$ and $\Gamma \vdash e : \tau_1 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$ then $\Gamma \vdash E[e] : \tau \mid \varepsilon$

Proof.

Induction on the evaluation context typing $\Gamma \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$.

- Case $E = \square$ trivial
- Case $E = E_0\ e_0$

$\Gamma \vdash_{\text{ec}} E_0\ e_0 : \tau_1 \to \tau \mid \varepsilon$	Given
$\Gamma \vdash_{\text{ec}} E_0 : \tau_1 \to \tau_3 \to^{\varepsilon} \tau \mid \varepsilon$	CAPP1
$\Gamma \vdash e_0 : \tau_3 \mid \varepsilon$	CAPP1
$\lceil E \rceil^l \triangleq \lceil E_0\ e_0 \rceil^l \triangleq \lceil E_0 \rceil^l$	
$\Gamma \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil^l, \varepsilon \rangle$	Given
$\Gamma \vdash E_0[e] : \tau_3 \to^{\varepsilon} \tau \mid \varepsilon$	I.H.
$\Gamma \vdash E_0[e]\ e_0 : \tau \mid \varepsilon$	T-E-App

- Case $E = \texttt{handle}^{\varepsilon}_m(h, E_0)$

$\Gamma \vdash_{\text{ec}} \texttt{handle}^{\varepsilon}_m(h, E_0) : \tau_1 \to \tau \mid \varepsilon$	Given
$\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$	CHANDLE

$\Gamma \vdash_{\text{ec}} E_0 : \tau_1 \to \tau \mid \langle l, \varepsilon \rangle$ CHANDLE
$\lceil E \rceil^l \triangleq \lceil \texttt{handle}^{\varepsilon}_m(h, E_0) \rceil^l \triangleq \langle l, \lceil E_0 \rceil^l \rangle = \langle \lceil E_0 \rceil^l, l \rangle$
$\Gamma \vdash e : \tau_1 \mid \langle \lceil E_0 \rceil^l, l, \varepsilon \rangle$ Given
$\Gamma \vdash E_0[e] : \tau_1 \to \tau \mid \langle l, \varepsilon \rangle$ I.H.
$\Gamma \vdash \texttt{handle}^{\varepsilon}_m(h, E_0[e]) : \tau \mid \varepsilon$ T-E-Handle

- Other cases follow similarly.

□

Lemma A.2. (Effect corresponds to the evaluation context)

If $\cdot \vdash E[e] : \tau \mid \varepsilon$ then there exists τ_1 such that $\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \varepsilon$ and $\cdot \vdash e : \tau_1 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$

Proof.

By induction on E.

- Case $E = \square$ trivial. $\tau_1 = \tau$
- Case $E = E_0\ e_0$

$\cdot \vdash (E_0\ e_0)[e] : \tau \mid \varepsilon$ Given
$\cdot \vdash E_0[e]\ e_0 : \tau \mid \varepsilon$
$\cdot \vdash E_0[e] : \tau_2 \to \tau \mid \varepsilon$ T-E-App
$\cdot \vdash e_0 : \tau_2 \mid \varepsilon$ T-E-App
$\cdot \vdash_{\text{ec}} E_0 : \tau_3 \to (\tau_2 \to \tau) \mid \varepsilon$ I.H.
$\cdot \vdash e : \tau_3 \mid \langle \lceil E_0 \rceil^l, \varepsilon \rangle$ I.H.
$\lceil E \rceil^l \triangleq \lceil E_0\ e_0 \rceil^l \triangleq \lceil E_0 \rceil^l$
$\cdot \vdash e : \tau_3 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$ I.H.
$\cdot \vdash_{\text{ec}} E : \tau_3 \to \tau \mid \varepsilon$ CAPP1
$\tau_1 = \tau_3$ Choose

- Case $E = \texttt{handle}^{\varepsilon}_m(h, E_0)$

$\cdot \vdash \texttt{handle}^{\varepsilon}_m(h, E_0)[e] : \tau \mid \varepsilon$ Given
$\cdot \vdash \texttt{handle}^{\varepsilon}_m(h, E_0[e]) : \tau \mid \varepsilon$
$\cdot \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$ T-E-Handle
$\cdot \vdash E_0[e] : \tau \mid \langle l, \varepsilon \rangle$ T-E-Handle
$\cdot \vdash_{\text{ec}} E_0 : \tau_2 \to \tau \mid \langle l, \varepsilon \rangle$ I.H.
$\cdot \vdash e : \tau_2 \mid \langle \lceil E_0 \rceil^l, l, \varepsilon \rangle$ I.H.
$\lceil E \rceil^l \triangleq \lceil \texttt{handle}^{\varepsilon}_m(h, E_0) \rceil^l \triangleq \langle l, \lceil E_0 \rceil^l \rangle = \langle \lceil E_0 \rceil^l, l \rangle$
$\cdot \vdash e : \tau_2 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$ I.H.

$\cdot \vdash_{\text{ec}} \texttt{handle}^{\varepsilon}_{m}(h, E_0) : \tau_2 \to \tau \mid \varepsilon$ — CHANDLE
$\tau_1 = \tau_2$ — Choose

- Other cases follow similarly.

□

Lemma A.3. (Well typed operations are handled)

If $\cdot \vdash E[\texttt{perform op}\ \overline{\tau}\ v] : \tau \mid \langle\rangle$ then E has the form $E_1 \cdot \texttt{handle}^{\varepsilon}_{m}(h, \cdot) \cdot E_2$ with $\texttt{op} \notin \texttt{bop}(E_2)$ and $\texttt{op} \to f \in h$.

Proof.

$\cdot \vdash E[\texttt{perform op}\ \overline{\tau}\ v] : \tau \mid \langle\rangle$ — Given
$\cdot \vdash \texttt{perform op}\ \overline{\tau}\ v : \tau_1 \mid \lceil E \rceil^{l}$ — Lemma A.2
$\cdot \vdash \texttt{perform op}\ \overline{\tau} : \tau_2 \to^{\lceil E \rceil^{l}} \tau_1 \mid \lceil E \rceil^{l}$ — T-E-App
$\cdot \vdash_{\text{val}} \texttt{perform op}\ \overline{\tau} : \tau_2 \to^{\lceil E \rceil^{l}} \tau_1$ — T-E-Val
$l \in \lceil E \rceil^{l}$ — $\lceil E \rceil^{l} = \langle l, \varepsilon' \rangle$
$E = E_1 \cdot \texttt{handle}^{\varepsilon}_{\text{d}}(h, \cdot) \cdot E_2$ — Definition of $\lceil E \rceil^{l}$
$\texttt{op} \to f \in h$
$\texttt{op} \notin \texttt{bop}(E_2)$ — Let $\texttt{handle}^{\varepsilon}_{\text{d}}(h, \cdot)$ be the inner most one

□

Lemma A.4. (Effects types are meaningful)

If $\cdot \vdash E[\texttt{perform op}\ \overline{\tau}\ v] : \tau \mid \varepsilon$ with $\texttt{op} \notin \texttt{bop}(E)$, then $\texttt{op} \in \Sigma(l)$ and $l \in \varepsilon$, i.e. effect types cannot be discarded without a handler.

Proof.

$\cdot \vdash E[\texttt{perform op}\ \overline{\tau}\ v] : \tau \mid \varepsilon$ — Given
$\cdot \vdash \texttt{perform op}\ \overline{\tau}\ v : \tau_1 \mid \langle \lceil E \rceil^{l}, \varepsilon \rangle$ — Lemma A.2
$\cdot \vdash \texttt{perform op}\ \overline{\tau} : \tau_2 \to^{\langle \lceil E \rceil^{l}, \varepsilon \rangle} \tau_1 \mid \langle \lceil E \rceil^{l}, \varepsilon \rangle$ — T-E-App
$\cdot \vdash_{\text{val}} \texttt{perform op}\ \overline{\tau} : \tau_2 \to^{\langle \lceil E \rceil^{l}, \varepsilon \rangle} \tau_1$ — T-E-Val
$l \in \langle \lceil E \rceil^{l}, \varepsilon \rangle$ — $\langle \lceil E \rceil^{l}, \varepsilon \rangle = \langle l, \varepsilon' \rangle$
$\texttt{op} \notin \texttt{bop}(E)$ — Given
$l \in \varepsilon$

□

A.2 Type Safety Theorem

Theorem A.5. (Effects Progress)

If $\cdot \vdash e_1 : \tau \mid \langle\rangle$ then e_1 is a value or $e_1 = \texttt{error}_p$ or $e_1 \Longrightarrow e_2$.

Proof. By Lemma A.6, e_1 is a value, or $e_1 = \texttt{error}_p$, or $e_1 \Longrightarrow e_2$, or $e_1 = E[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$ with $\texttt{op} \notin \texttt{bop}(E)$ and $\texttt{op} \in \Sigma(l)$. The last case is impossible:

$\cdot \vdash E[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v] : \tau \mid \langle\rangle$	Given
$l \in \langle\rangle$	Lemma A.4
Contradiction	$\langle\rangle$ is empty

□

Lemma A.6. (`Effects` Progress with effects)

If $\cdot \vdash e_1 : \tau \mid \varepsilon$ then either

- e_1 is a value or
- $e_1 = \texttt{error}_p$ or
- $e_1 \Longrightarrow e_2$ or
- $e_1 = E[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$ where $\texttt{op} : \forall\overline{\alpha}.\tau_1 \to \tau_2 \in \Sigma(l)$, and $\texttt{op} \notin \texttt{bop}(E)$.

Proof. By induction on $\cdot \vdash e_1 : \tau \mid \varepsilon$

- Case $e_1 = v$, trivially true
- Case $e_1 = \texttt{error}_p$, trivially true
- Case $e_1 = e_3\ e_4$

$\cdot \vdash e_3\ e_4 : \tau \mid \varepsilon$	Given
$\cdot \vdash e_3 : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$	Given
$\cdot \vdash e_4 : \tau_1 \mid \varepsilon$	Given

  By I.H. on e_3:

  - Case $e_3 \Longrightarrow e_5$

    $e_1 = e_3\ e_4 \Longrightarrow e_5\ e_4 = e_2$

  - Case $e_3 = \texttt{error}_p$

    $e_1 = \texttt{error}_p\ e_4 \Longrightarrow \texttt{error}_p = e_2$

  - Case $e_3 = E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$ where $\texttt{op} : \forall\overline{\alpha}.\tau_1 \to \tau_2 \in \Sigma(l)$, and $\texttt{op} \notin \texttt{bop}(E)$

    $e_1 = E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]\ e_4 = E_1[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$.

  - Case e_3 is a value

    By I.H. on e_4.

    – Cases $e_4 \Longrightarrow e_5$, $e_4 = \texttt{error}_p$ makes e_1 steps.

    – Case e_4 is a value, by canonical forms of e_3, only some cases are valid:

      - Case $e_3 = \lambda x.e_6$. Then $e_1 = (\lambda x.e_6)\ e_4 \Longrightarrow e_6[x := e_4] = e_2$

  - Case $e_3 = \texttt{perform}\ \texttt{op}\ \overline{\tau}$. Then $e_1 = \square[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ e_4]$
  - Case $e_3 = \texttt{handler}^{\varepsilon}_{m}(h)$. Then $e_1 = \texttt{handler}^{\varepsilon}_{m}(h)\ e_4 \Longrightarrow \texttt{handle}^{\varepsilon}_{m}(h, e_4\ ()) = e_2$

- Case $e_1 = e_3\ [\tau_1]$

  $\cdot \vdash e_3\ [\tau_1] : \tau_2[\alpha := \tau_1] \mid \varepsilon$ Given
  $\cdot \vdash e_3 : \forall \alpha.\tau_2 \mid \varepsilon$

  By I.H. on e_3:

  - Case $e_3 \Longrightarrow e_5$

    $e_1 = e_3\ [\tau_1] \Longrightarrow e_5\ [\tau_1] = e_2$

  - Case $e_3 = \texttt{error}_p$

    $e_1 = \texttt{error}_p\ [\tau_1] \Longrightarrow \texttt{error}_p = e_2$

  - Case $e_3 = E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$ where $\texttt{op} : \forall \overline{\alpha}.\tau_1 \to \tau_2 \in \Sigma(l)$, and $\texttt{op} \notin \texttt{bop}(E)$

    $e_1 = E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]\ [\tau_1] = E_1[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$.

  - Case e_3 is a value

    By canonical form of value, $e_3 = \Lambda\alpha.e_4$

    $e_1 = (\Lambda\alpha.e_4)\ [\tau_1] \Longrightarrow e_4[\alpha := \tau_1] = e_2$

- Case $e_1 = \texttt{handle}^{\varepsilon}_{m}(h, e_3)$

  $\cdot \vdash \texttt{handle}^{\varepsilon}_{m}(h, e_3) : \tau \mid \varepsilon$ Given
  $\cdot \vdash e_3 : \tau \mid \langle l, \varepsilon \rangle$ Given

  By I.H. on e_3:

  - Case $e_3 \Longrightarrow e_4$

    $e_1 = \texttt{handle}^{\varepsilon}_{m}(h, e_3) \Longrightarrow \texttt{handle}^{\varepsilon}_{m}(h, e_4) = e_2$

  - Case $e_3 = \texttt{error}_p$

    $e_1 = \texttt{handle}^{\varepsilon}_{m}(h, \texttt{error}_p) \Longrightarrow \texttt{error}_p = e_2$

  - Case $e_3 = E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v]$ where $\texttt{op} : \forall \overline{\alpha}.\tau_1 \to \tau_2 \in \Sigma(l)$, and $\texttt{op} \notin \texttt{bop}(E)$

    $e_1 = \texttt{handle}^{\varepsilon}_{m}(h, E_0[\texttt{perform}\ \texttt{op}\ \overline{\tau}\ v])$.

    – Case $\texttt{op} \to f \in h$

      - Case $m = s$ shallow handler

        $e_1 \Longrightarrow f\ \overline{\tau}\ v\ (\lambda x.E_0[x]) = e_2$

      - Case $m = d$ deep handler

        $e_1 \Longrightarrow f\ \overline{\tau}\ v\ (\lambda x.\texttt{handle}^{\varepsilon}_{d}(h, E_0[x])) = e_2$

  - Case $\mathsf{op} \to f \notin h$

    $$e_1 = E_1[\texttt{perform}\ \mathsf{op}\ \overline{\tau}\ v].$$

  - Case e_3 is a value

    $$e_1 = \texttt{handle}^{\varepsilon}_{m}(h, e_3) \Longrightarrow e_3 = e_2$$

- Other cases follow similarly.

□

Theorem A.7. (`Effects` Preservation)

If $\cdot \vdash e_1 : \tau \mid \langle\rangle$ and $e_1 \Longrightarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \langle\rangle$.

Proof. By cases on the step.

- Case $e_1 = E[e_3] \Longrightarrow E[e_4] = e_2$ and $e_3 \rightarrow e_4$.

$\cdot \vdash E[e_3] : \tau \mid \langle\rangle$	Given
$\cdot \vdash_{\text{ec}} E : \tau_1 \to \tau \mid \langle\rangle$	Lemma A.2
$\cdot \vdash e_3 : \tau_1 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$	Lemma A.2
$\cdot \vdash e_4 : \tau_1 \mid \langle \lceil E \rceil^l, \varepsilon \rangle$	Lemma A.8
$\cdot \vdash E[e_4] : \tau \mid \langle\rangle$	Lemma A.1

- Case Step-E-Error, $E\left[\texttt{error}_p\right] \Longrightarrow \texttt{error}_p$.

$\cdot \vdash_{\text{val}} \texttt{error}_p : \tau$	T-E-Error
$\cdot \vdash \texttt{error}_p : \tau \mid \varepsilon$	T-E-Val

□

Lemma A.8. (Redex preservation)

If $\cdot \vdash e_1 : \tau \mid \varepsilon$ and $e_1 \rightarrow e_2$ then $\cdot \vdash e_2 : \tau \mid \varepsilon$.

Proof. By cases on $e_1 \rightarrow e_2$.

- Case $(\lambda^{\varepsilon} x : \tau_1.e)\ v \rightarrow e[x := v]$.

$\cdot \vdash \lambda^{\varepsilon} x : \tau_1.e : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$	T-E-App
$\cdot \vdash v : \tau_1 \mid \varepsilon$	T-E-App
$x : \tau_1 \vdash e : \tau \mid \varepsilon$	T-E-Val, T-E-Abs
$\cdot \vdash_{\text{val}} v : \tau_1$	T-E-Val
$\cdot \vdash e[x := v] : \tau \mid \varepsilon$	Lemma A.9

- Case $\left(\Lambda\alpha^k.v\right)[\tau_0] \rightarrow v[\alpha := \tau_0]$

$\cdot \vdash \left(\Lambda\alpha^k.v\right)[\tau_0] : \tau_1[\alpha := \tau_0] \mid \varepsilon$	Given
$\cdot \vdash \Lambda\alpha^k.v : \forall\alpha^k.\tau_1 \mid \varepsilon$	T-E-Tapp

$\vdash_{\text{wf}} \tau_0 : k$ T-E-Tapp

$\cdot \vdash_{\text{val}} v : \tau_1$ T-E-Val, T-E-TAbs

$\cdot \vdash_{\text{val}} v[\alpha := \tau_0] : \tau_1[\alpha := \tau_0]$ Lemma A.10

$\cdot \vdash v[\alpha := \tau_0] : \tau \mid \varepsilon$ T-E-Val

- Case $(\mu x : \tau_1.v_1)\, v_2 \rightarrow v_1[x := \mu x : \tau_1.v_1]\, v_2$, where $\tau_1 = \tau_2 \to^{\varepsilon} \tau$.

$\cdot \vdash (\mu x : \tau_1.v_1)\, v_2 : \tau \mid \varepsilon$ Given

$\cdot \vdash \mu x : \tau_1.v_1 : \tau_1 \mid \varepsilon$ T-E-App

$\cdot \vdash v_2 : \tau_2 \mid \varepsilon$ T-E-App

$\cdot \vdash_{\text{val}} \mu x : \tau_1.v_1 : \tau_1$ T-E-Val

$x : \tau_1 \vdash_{\text{val}} v_1 : \tau_1$ T-E-Fix

$\cdot \vdash_{\text{val}} v_1[x := \mu x : \tau_1.v_1] : \tau_1$ Lemma A.9

$\cdot \vdash v_1[x := \mu x : \tau_1.v_1] : \tau_1 \mid \varepsilon$ T-E-Val

$\cdot \vdash v_1[x := \mu x : \tau_1.v_1]\, v_2 : \tau \mid \varepsilon$ T-E-App

- Case $(\mu x : \tau_1.v)[\tau_2] \rightarrow (v[x := \mu x : \tau_1.v])[\tau_2]$, where $\tau_1 = \forall \alpha^k.\tau_3$.

$\cdot \vdash (\mu x : \tau_1.v)[\tau_2] : \tau_3[\alpha := \tau_2] \mid \varepsilon$ Given

$\vdash_{\text{wf}} \tau_2 : k$ T-E-Tapp

$\cdot \vdash \mu x : \tau_1.v : \tau_1 \mid \varepsilon$ T-E-Tapp

$\cdot \vdash_{\text{val}} \mu x : \tau_1.v : \tau_1$ T-E-Val

$x : \tau_1 \vdash_{\text{val}} v : \tau_1$ T-E-Fix

$\cdot \vdash_{\text{val}} v[x := \mu x : \tau_1.v] : \tau_1$ Lemma A.9

$\cdot \vdash v[x := \mu x : \tau_1.v] : \tau_1 \mid \varepsilon$ T-E-Val

$\cdot \vdash (v[x := \mu x : \tau_1.v])[\tau_2] : \tau_3[\alpha := \tau_2] \mid \varepsilon$ T-E-Tapp

- Case $\texttt{handler}^{\varepsilon}_{m}(h)\, v \rightarrow \texttt{handle}^{\varepsilon}_{m}(h, v\ \mathsf{unit})$.

$\cdot \vdash \texttt{handler}^{\varepsilon}_{m}(h)\, v : \tau \mid \varepsilon$ Given

$\cdot \vdash \texttt{handler}^{\varepsilon}_{m}(h) : \tau_1 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App

$\cdot \vdash v : \tau_1 \mid \varepsilon$ T-E-App

$\cdot \vdash_{\text{val}} \texttt{handler}^{\varepsilon}_{m}(h) : \tau_1 \to^{\varepsilon} \tau$ T-E-Val

$\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ T-E-Handler

$\tau_1 = \mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau$ T-E-Handler

$\cdot \vdash v : \mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau \mid \langle l \mid \varepsilon \rangle$ Lemma A.11

$\cdot \vdash v\ \mathsf{unit} : \tau \mid \langle l \mid \varepsilon \rangle$ T-Unit, T-E-App

$\cdot \vdash \texttt{handle}^{\varepsilon}_{m}(h, v\ \mathsf{unit}) : \tau \mid \varepsilon$ T-E-Handle

- Case $\texttt{handle}^{\varepsilon}_{m}(h, v) \rightarrow v$.

$\cdot \vdash v : \tau \mid \langle l \mid \varepsilon \rangle$ T-E-Handle

$\cdot \vdash_{\text{val}} v : \tau$ T-E-Val
$\cdot \vdash v : \tau \mid \varepsilon$ Lemma A.11

- Case $\texttt{handle}^{\varepsilon}_{d}(h, E[\texttt{perform op } \tau_0\ v]) \rightarrow f[\tau_0]\ v\ k$, where $\texttt{op} \notin \texttt{bop}(E)$, $(\texttt{op} \to f) \in h$, $\texttt{op} : \forall\alpha.\tau_1 \to \tau_2 \in \Sigma(l)$ and $k = \lambda^{\varepsilon} x : \tau_2[\alpha := \tau_0].\texttt{handle}^{\varepsilon}_{d}(h, E[x])$.

$\cdot \vdash \texttt{handle}^{\varepsilon}_{d}(h, E[\texttt{perform op } \tau_0\ v]) : \tau \mid \varepsilon$ Given
$\cdot \vdash_{\text{ops}} h : \tau \mid l \mid \varepsilon$ T-E-Handle
$\cdot \vdash f : \forall\alpha.\tau_1 \to^{\varepsilon} ((\tau_2 \to^{\varepsilon} \tau) \to^{\varepsilon} \tau) \mid \varepsilon$ T-E-Ops
$\cdot \vdash_{\text{val}} f : \forall\alpha.\tau_1 \to^{\varepsilon} ((\tau_2 \to^{\varepsilon} \tau) \to^{\varepsilon} \tau)$ T-E-Val
$\cdot \vdash f[\tau_0] : \tau_1[\alpha := \tau_0] \to^{\varepsilon} ((\tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau) \to^{\varepsilon} \tau) \mid \varepsilon$ T-E-Val, T-E-Tapp
$\cdot \vdash \texttt{perform op } \tau_0\ v : \tau_2[\alpha := \tau_0] \mid \langle \lceil \texttt{handle}^{\varepsilon}_{d}(h, E) \rceil^{l}, \varepsilon \rangle$ Lemma A.2
$\cdot \vdash_{\text{ec}} \texttt{handle}^{\varepsilon}_{d}(h, E) : \tau_2[\alpha := \tau_0] \to \tau \mid \varepsilon$ Lemma A.2
$\cdot \vdash v : \tau_1[\alpha := \tau_0] \mid \varepsilon$ T-E-App, Lemma A.11
$\cdot \vdash f[\tau_0]\ v : (\tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau) \to^{\varepsilon} \tau \mid \varepsilon$ T-E-Val, T-E-Tapp

build up $k \downarrow$

$x : \tau_2[\alpha := \tau_0] \vdash_{\text{val}} x : \tau_2[\alpha := \tau_0]$ T-E-Var
$x : \tau_2[\alpha := \tau_0] \vdash x : \tau_2[\alpha := \tau_0] \mid \varepsilon$ T-E-Val
$x : \tau_2[\alpha := \tau_0] \vdash_{\text{ec}} \texttt{handle}^{\varepsilon}_{d}(h, E) : \tau_2[\alpha := \tau_0] \to \tau \mid \varepsilon$ weakening
$x : \tau_2[\alpha := \tau_0] \vdash \texttt{handle}^{\varepsilon}_{d}(h, E[x]) : \tau \mid \varepsilon$ Lemma A.1
$\cdot \vdash_{\text{val}} \lambda^{\varepsilon} x : \tau_2[\alpha := \tau_0].\texttt{handle}^{\varepsilon}_{d}(h, E[x]) : \tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau$ T-E-Abs
$\cdot \vdash \lambda^{\varepsilon} x : \tau_2[\alpha := \tau_0].\texttt{handle}^{\varepsilon}_{d}(h, E[x]) : \tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau \mid \varepsilon$ T-E-Val
$\cdot \vdash k : \tau_2[\alpha := \tau_0] \to^{\varepsilon} \tau \mid \varepsilon$ T-E-Abs, T-E-Val
$\cdot \vdash f[\tau_0]\ v\ k : \tau \mid \varepsilon$ T-E-App

- Case $\texttt{handle}^{\varepsilon}_{s}(h, E[\texttt{perform op } \tau_0\ v]) \rightarrow f[\tau_0]\ v\ k$, where $\texttt{op} \notin \texttt{bop}(E)$, $(\texttt{op} \to f) \in h$, $\texttt{op} : \forall\alpha.\tau_1 \to \tau_2 \in \Sigma(l)$ and $k = \lambda^{\langle l, \varepsilon \rangle} x : \tau_2[\alpha := \tau_0].E[x]$, with x fresh.

$\cdot \vdash \texttt{handle}^{\varepsilon}_{s}(h, E[\texttt{perform op } \tau_0\ v]) : \tau \mid \varepsilon$ Given
$\cdot \vdash_{\text{ops}} s : h : \tau \mid l \mid \varepsilon$ T-E-Handle
$\cdot \vdash E[\texttt{perform op } \tau_0\ v] : \tau \mid \langle l, \varepsilon \rangle$ T-E-Handle
$\cdot \vdash_{\text{val}} f : \forall\alpha.\tau_1 \to^{\varepsilon} (\tau_2 \to^{\langle l, \varepsilon \rangle} \tau) \to^{\varepsilon} \tau$ T-E-Ops-Shallow
$\cdot \vdash f[\tau_0] : \tau_1[\alpha := \tau_0] \to^{\varepsilon} (\tau_2[\alpha := \tau_0] \to^{\langle l, \varepsilon \rangle} \tau) \to^{\varepsilon} \tau \mid \varepsilon$ T-E-Val, T-E-Tapp
$\cdot \vdash \texttt{perform op } \tau_0\ v : \tau_2[\alpha := \tau_0] \mid \langle \lceil E \rceil^{l}, \langle l, \varepsilon \rangle \rangle$ Lemma A.2
$\cdot \vdash_{\text{ec}} E : \tau_2[\alpha := \tau_0] \to \tau \mid \langle l, \varepsilon \rangle$ Lemma A.2
$\cdot \vdash v : \tau_1[\alpha := \tau_0] \mid \varepsilon$ T-E-App, Lemma A.11
$\cdot \vdash f[\tau_0]\ v : (\tau_2[\alpha := \tau_0] \to^{\langle l, \varepsilon \rangle} \tau) \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App
$x : \tau_2[\alpha := \tau_0] \vdash_{\text{val}} x : \tau_2[\alpha := \tau_0]$ T-E-Var
$x : \tau_2[\alpha := \tau_0] \vdash x : \tau_2[\alpha := \tau_0] \mid \langle \lceil E \rceil^{l}, \langle l, \varepsilon \rangle \rangle$ T-E-Val
$x : \tau_2[\alpha := \tau_0] \vdash_{\text{ec}} E : \tau_2[\alpha := \tau_0] \to \tau \mid \langle l, \varepsilon \rangle$ Weakening

$x : \tau_2[\alpha := \tau_0] \vdash E[x] : \tau \mid \langle l, \varepsilon \rangle$ — Lemma A.1
$\cdot \vdash_{\text{val}} k : \tau_2[\alpha := \tau_0] \to^{\langle l, \varepsilon \rangle} \tau$ — T-E-Abs
$\cdot \vdash k : \tau_2[\alpha := \tau_0] \to^{\langle l, \varepsilon \rangle} \tau \mid \varepsilon$ — T-E-Val
$\cdot \vdash f[\tau_0]\, v\, k : \tau \mid \varepsilon$ — T-E-App

- Other cases follows similarly to the above and [?], with every judgment under effect ε.

□

The following proofs assume capture-avoiding substitution, weakening, and well-scoped type abstraction. The freshness condition for T-E-TAbs is not yet shown in the rule.

Lemma A.9. (Variable Substitution)

If $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v : \tau_1$ then the following hold:

- If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e : \tau \mid \varepsilon$, then $\Gamma_1, \Gamma_2 \vdash e[x := v] : \tau \mid \varepsilon$.
- If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} v_0 : \tau$, then $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v_0[x := v] : \tau$.
- If $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$, then $\Gamma_1, \Gamma_2 \vdash_{\text{ops}} m : h[x := v] : \tau \mid l \mid \varepsilon$.

Proof. By mutual induction on typing, following Lemma 18 of [?]. Rename bound variables fresh as needed.

- Case $v_0 = x$.

  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} x : \tau_1$ — Given
  $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v : \tau_1$ — Given
  $\Gamma_1, \Gamma_2 \vdash_{\text{val}} x[x := v] : \tau_1$ — Substitution

- Case $v_0 = y$, where $y \neq x$.

  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} y : \tau$ — Given
  $y : \tau \in \Gamma_1, \Gamma_2$ — T-E-Var
  $\Gamma_1, \Gamma_2 \vdash_{\text{val}} y[x := v] : \tau$ — T-E-Var

- Case $e = v_0$.

  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash v_0 : \tau \mid \varepsilon$ — Given
  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} v_0 : \tau$ — T-E-Val
  $\Gamma_1, \Gamma_2 \vdash_{\text{val}} v_0[x := v] : \tau$ — I.H.
  $\Gamma_1, \Gamma_2 \vdash v_0[x := v] : \tau \mid \varepsilon$ — T-E-Val

- Case $e = e_1\, e_2$.

  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_1\, e_2 : \tau \mid \varepsilon$ — Given
  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_1 : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ — T-E-App
  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_2 : \tau_2 \mid \varepsilon$ — T-E-App

$\Gamma_1, \Gamma_2 \vdash e_1[x := v] : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ — I.H.
$\Gamma_1, \Gamma_2 \vdash e_2[x := v] : \tau_2 \mid \varepsilon$ — I.H.
$\Gamma_1, \Gamma_2 \vdash e_1[x := v]\ e_2[x := v] : \tau \mid \varepsilon$ — T-E-App

- Case $e = e_1[\tau_2]$.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_1[\tau_2] : \tau[\alpha := \tau_2] \mid \varepsilon$ — Given
$\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_1 : \forall \alpha^k . \tau \mid \varepsilon$ — T-E-Tapp
$\vdash_{\text{wf}} \tau_2 : k$ — T-E-Tapp
$\Gamma_1, \Gamma_2 \vdash e_1[x := v] : \forall \alpha^k . \tau \mid \varepsilon$ — I.H.
$\Gamma_1, \Gamma_2 \vdash (e_1[x := v])[\tau_2] : \tau[\alpha := \tau_2] \mid \varepsilon$ — T-E-Tapp

- Case $v_0 = \lambda^{\varepsilon} y : \tau_2 . e$. Choose y fresh for v.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} \lambda^{\varepsilon} y : \tau_2 . e : \tau_2 \to^{\varepsilon} \tau$ — Given
$\Gamma_1, x : \tau_1, \Gamma_2, y : \tau_2 \vdash e : \tau \mid \varepsilon$ — T-E-Abs
$\Gamma_1, \Gamma_2, y : \tau_2 \vdash_{\text{val}} v : \tau_1$ — Given, weakening
$\Gamma_1, \Gamma_2, y : \tau_2 \vdash e[x := v] : \tau \mid \varepsilon$ — I.H.
$\Gamma_1, \Gamma_2 \vdash_{\text{val}} \lambda^{\varepsilon} y : \tau_2 . e[x := v] : \tau_2 \to^{\varepsilon} \tau$ — T-E-Abs

- Case $v_0 = \Lambda \alpha^k . v_1$. Choose α fresh for the context and v.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} \Lambda \alpha^k . v_1 : \forall \alpha^k . \tau$ — Given
$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} v_1 : \tau$ — T-E-TAbs
$\Gamma_1, \Gamma_2 \vdash_{\text{val}} v_1[x := v] : \tau$ — I.H.
$\Gamma_1, \Gamma_2 \vdash_{\text{val}} \Lambda \alpha^k . v_1[x := v] : \forall \alpha^k . \tau$ — T-E-TAbs

- Case $v_0 = \mu y : \tau_2 . v_1$. Choose y fresh for v.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} \mu y : \tau_2 . v_1 : \tau$ — Given
$\Gamma_1, x : \tau_1, \Gamma_2, y : \tau_2 \vdash_{\text{val}} v_1 : \tau$ — T-E-Fix
$\Gamma_1, \Gamma_2, y : \tau_2 \vdash_{\text{val}} v : \tau_1$ — Given, weakening
$\Gamma_1, \Gamma_2, y : \tau_2 \vdash_{\text{val}} v_1[x := v] : \tau$ — I.H.
$\Gamma_1, \Gamma_2 \vdash_{\text{val}} \mu y : \tau_2 . v_1[x := v] : \tau$ — T-E-Fix

- Case $h = \{\mathsf{op}_1 \to f_1, ..., \mathsf{op}_n \to f_n\}$.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{ops}} m : \{\mathsf{op}_1 \to f_1, ..., \mathsf{op}_n \to f_n\} : \tau \mid l \mid \varepsilon$ — Given
$\Gamma_1, \Gamma_2 \vdash_{\text{ops}} m : \{\mathsf{op}_1 \to f_1[x := v], ..., \mathsf{op}_n \to f_n[x := v]\} : \tau \mid l \mid \varepsilon$ — I.H.

- Case $v_0 = \mathtt{handler}^{\varepsilon}_m(h)$.

$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{val}} \mathtt{handler}^{\varepsilon}_m(h) : \left(\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau\right) \to^{\varepsilon} \tau$ — Given
$\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$ — T-E-Handler
$\Gamma_1, \Gamma_2 \vdash_{\text{ops}} m : h[x := v] : \tau \mid l \mid \varepsilon$ — I.H.

$\Gamma_1, \Gamma_2 \vdash_{\text{val}} \texttt{handler}^{\varepsilon}_{m}(h[x := v]) : \left(\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau\right) \to^{\varepsilon} \tau$ T-E-Handler

- Case $e = \texttt{handle}^{\varepsilon}_{m}(h, e_0)$.

  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash \texttt{handle}^{\varepsilon}_{m}(h, e_0) : \tau \mid \varepsilon$ Given
  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$ T-E-Handle
  $\Gamma_1, x : \tau_1, \Gamma_2 \vdash e_0 : \tau \mid \langle l \mid \varepsilon \rangle$ T-E-Handle
  $\Gamma_1, \Gamma_2 \vdash_{\text{ops}} m : h[x := v] : \tau \mid l \mid \varepsilon$ I.H.
  $\Gamma_1, \Gamma_2 \vdash e_0[x := v] : \tau \mid \langle l \mid \varepsilon \rangle$ I.H.
  $\Gamma_1, \Gamma_2 \vdash \texttt{handle}^{\varepsilon}_{m}(h[x := v], e_0[x := v]) : \tau \mid \varepsilon$ T-E-Handle

- Other cases follow similarly.

□

Lemma A.10. (Type Variable Substitution)

If $\vdash_{\text{wf}} \tau_1 : k_1$, then the following hold:

- If $\Gamma \vdash e : \tau \mid \varepsilon$, then $\Gamma[\alpha := \tau_1] \vdash e[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$.
- If $\Gamma \vdash_{\text{val}} v : \tau$, then $\Gamma[\alpha := \tau_1] \vdash_{\text{val}} v[\alpha := \tau_1] : \tau[\alpha := \tau_1]$.
- If $\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$, then $\Gamma[\alpha := \tau_1] \vdash_{\text{ops}} m : h[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid l \mid \varepsilon[\alpha := \tau_1]$.

Proof. By mutual induction on typing, following Lemma 20 of [?]. Rename bound variables fresh as needed; effect labels are constants and operation signatures are closed except for their bound variables.

- Case $e = v$.

  $\Gamma \vdash v : \tau \mid \varepsilon$ Given
  $\Gamma \vdash_{\text{val}} v : \tau$ T-E-Val
  $\Gamma[\alpha := \tau_1] \vdash_{\text{val}} v[\alpha := \tau_1] : \tau[\alpha := \tau_1]$ I.H.
  $\Gamma[\alpha := \tau_1] \vdash v[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ T-E-Val

- Case $e = e_1\ e_2$.

  $\Gamma \vdash e_1\ e_2 : \tau \mid \varepsilon$ Given
  $\Gamma \vdash e_1 : \tau_2 \to^{\varepsilon} \tau \mid \varepsilon$ T-E-App
  $\Gamma \vdash e_2 : \tau_2 \mid \varepsilon$ T-E-App
  $\Gamma[\alpha := \tau_1] \vdash e_1[\alpha := \tau_1] : (\tau_2 \to^{\varepsilon} \tau)[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ I.H.
  $\Gamma[\alpha := \tau_1] \vdash e_2[\alpha := \tau_1] : \tau_2[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ I.H.
  $\Gamma[\alpha := \tau_1] \vdash e_1[\alpha := \tau_1]\ e_2[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ T-E-App

- Case $e = e_1[\tau_2]$. Choose β fresh for α and τ_1.

  $\Gamma \vdash e_1[\tau_2] : \tau[\beta := \tau_2] \mid \varepsilon$ Given

$\Gamma \vdash e_1 : \forall \beta^k.\tau \mid \varepsilon$ — T-E-Tapp

$\vdash_{\text{wf}} \tau_2 : k$ — T-E-Tapp

$\Gamma[\alpha := \tau_1] \vdash e_1[\alpha := \tau_1] : \forall \beta^k.\tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ — I.H.

$\vdash_{\text{wf}} \tau_2[\alpha := \tau_1] : k$ — Kind substitution

$\Gamma[\alpha := \tau_1] \vdash (e_1[\alpha := \tau_1])[\tau_2[\alpha := \tau_1]] : (\tau[\alpha := \tau_1])[\beta := \tau_2[\alpha := \tau_1]] \mid \varepsilon[\alpha := \tau_1]$ — T-E-Tapp

$\Gamma[\alpha := \tau_1] \vdash (e_1[\tau_2])[\alpha := \tau_1] : (\tau[\beta := \tau_2])[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ — Substitution composition

- Case $v = \lambda^{\varepsilon} y : \tau_2.e$.

$\Gamma \vdash_{\text{val}} \lambda^{\varepsilon} y : \tau_2.e : \tau_2 \to^{\varepsilon} \tau$ — Given

$\Gamma, y : \tau_2 \vdash e : \tau \mid \varepsilon$ — T-E-Abs

$\Gamma[\alpha := \tau_1], y : \tau_2[\alpha := \tau_1] \vdash e[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ — I.H.

$\Gamma[\alpha := \tau_1] \vdash_{\text{val}} (\lambda^{\varepsilon} y : \tau_2.e)[\alpha := \tau_1] : (\tau_2 \to^{\varepsilon} \tau)[\alpha := \tau_1]$ — T-E-Abs

- Case $v = \Lambda \beta^k.v_0$. Choose $\beta \neq \alpha$ fresh for τ_1 and Γ.

$\Gamma \vdash_{\text{val}} \Lambda \beta^k.v_0 : \forall \beta^k.\tau$ — Given

$\Gamma \vdash_{\text{val}} v_0 : \tau$ — T-E-TAbs

$\Gamma[\alpha := \tau_1] \vdash_{\text{val}} v_0[\alpha := \tau_1] : \tau[\alpha := \tau_1]$ — I.H.

$\Gamma[\alpha := \tau_1] \vdash_{\text{val}} \big(\Lambda \beta^k.v_0\big)[\alpha := \tau_1] : \big(\forall \beta^k.\tau\big)[\alpha := \tau_1]$ — T-E-TAbs

- Case $v = \mu y : \tau_2.v_0$.

$\Gamma \vdash_{\text{val}} \mu y : \tau_2.v_0 : \tau$ — Given

$\Gamma, y : \tau_2 \vdash_{\text{val}} v_0 : \tau$ — T-E-Fix

$\Gamma[\alpha := \tau_1], y : \tau_2[\alpha := \tau_1] \vdash_{\text{val}} v_0[\alpha := \tau_1] : \tau[\alpha := \tau_1]$ — I.H.

$\Gamma[\alpha := \tau_1] \vdash_{\text{val}} (\mu y : \tau_2.v_0)[\alpha := \tau_1] : \tau[\alpha := \tau_1]$ — T-E-Fix

- Case $h = \{\mathtt{op}_1 \to f_1, ..., \mathtt{op}_n \to f_n\}$. Choose the signature-bound variables fresh.

$\Gamma \vdash_{\text{ops}} m : \{\mathtt{op}_1 \to f_1, ..., \mathtt{op}_n \to f_n\} : \tau \mid l \mid \varepsilon$ — Given

$\Gamma[\alpha := \tau_1] \vdash_{\text{ops}} m : \left\{\mathtt{op}_1 \to f_1[\alpha := \tau_1], ..., \mathtt{op}_n \to f_{n[\alpha := \tau_1]}\right\} : \tau[\alpha := \tau_1] \mid l \mid \varepsilon[\alpha := \tau_1]$ — I.H.

- Case $v = \mathtt{handler}^{\varepsilon}_m(h)$.

$\Gamma \vdash_{\text{val}} \mathtt{handler}^{\varepsilon}_m(h) : \big(\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau\big) \to^{\varepsilon} \tau$ — Given

$\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$ — T-E-Handler

$\Gamma[\alpha := \tau_1] \vdash_{\text{ops}} m : h[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid l \mid \varepsilon[\alpha := \tau_1]$ — I.H.

$\Gamma[\alpha := \tau_1] \vdash_{\text{val}} \mathtt{handler}^{\varepsilon[\alpha := \tau_1]}_m(h[\alpha := \tau_1]) : \big(\big(\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau\big) \to^{\varepsilon} \tau\big)[\alpha := \tau_1]$ — T-E-Handler

- Case $e = \texttt{handle}^{\varepsilon}_{m}(h, e_0)$.

  $\Gamma \vdash \texttt{handle}^{\varepsilon}_{m}(h, e_0) : \tau \mid \varepsilon$ — Given
  $\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon$ — T-E-Handle
  $\Gamma \vdash e_0 : \tau \mid \langle l \mid \varepsilon \rangle$ — T-E-Handle
  $\Gamma[\alpha := \tau_1] \vdash_{\text{ops}} m : h[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid l \mid \varepsilon[\alpha := \tau_1]$ — I.H.
  $\Gamma[\alpha := \tau_1] \vdash e_0[\alpha := \tau_1] : \tau[\alpha := \tau_1] \mid \langle l \mid \varepsilon[\alpha := \tau_1] \rangle$ — I.H.
  $\Gamma[\alpha := \tau_1] \vdash \texttt{handle}^{\varepsilon[\alpha:=\tau_1]}_{m}(h[\alpha := \tau_1], e_0[\alpha := \tau_1]) : \tau[\alpha := \tau_1] \mid \varepsilon[\alpha := \tau_1]$ — T-E-Handle

- Other cases follow similarly, using kind preservation and type-substitution composition.

□

Lemma A.11. (Values take any effect)

If $\Gamma \vdash_{\text{val}} v : \tau$ then $\Gamma \vdash v : \tau \mid \varepsilon$, for any well-formed effect ε.

Proof.

$\Gamma \vdash_{\text{val}} v : \tau$ — Given
$\Gamma \vdash v : \tau \mid \varepsilon$ — T-E-Val

□